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GEOGRAPHY

Fundamentals of Physical Geography

Class 6

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Chapter 1

Geography as a Discipline

Geography is the study of the earth's surface, its physical features, climate, soils, vegetation and the human activities that shape and are shaped by it. It bridges the natural and social sciences. Physical geography examines landforms, climate, water and biosphere, while human geography studies population, settlements, economic activities and cultural patterns. The discipline employs tools such as maps, remote sensing, GIS and field observation to understand spatial patterns and processes. Understanding geography is essential for resource management, disaster mitigation and sustainable development.

Chapter 2

The Origin and Evolution of the Earth

The earth formed approximately 4.6 billion years ago from the solar nebula. The Big Bang theory describes the origin of the universe around 13.8 billion years ago. The early earth was hot and molten; gradual cooling formed the crust. Theories of continental drift by Wegener and seafloor spreading led to plate tectonics, which explains the movement of lithospheric plates and the formation of mountains, trenches and volcanic arcs. The evolution of the atmosphere and the origin of life in the oceans marked critical milestones in making earth habitable.

Chapter 3

Climate

Climate refers to the average weather conditions of a place over a long period. The earth's climate is driven by insolation, atmospheric circulation, ocean currents and the distribution of land and water. The Köppen classification groups climates into tropical, dry, temperate, continental and polar types. Monsoons, the seasonal reversal of winds, profoundly influence the Indian subcontinent. Factors such as latitude, altitude, pressure belts and the Coriolis effect shape global climate patterns. Human-induced climate change, caused by greenhouse gas emissions, is altering temperature and precipitation regimes worldwide.